Oorjith Komarraju

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Research interests: Photonics, RF systems, and semiconductor devices

EDUCATION

University of California, San Diego **Expected 2027**

B.S. Electrical Engineering, Controls and Machine Learning *La Jolla, CA*

Relevant coursework: Circuit analysis and design; linear systems; digital design and RTL; modern physics; electromagnetic fields and transmission lines (in progress).

ENGINEERING EXPERIENCE

Triton Robotics **October 2025 - Present**

Electronics Engineer *La Jolla, CA*

- Design and validate custom electronic circuits in EasyEDA through simulation and physical testing.
- Develop a robot testbench power-module schematic with 24 V distribution and planned 12 V and 5 V outputs; evaluate output fusing and protection against motor-generated back current.
- Develop C++ control algorithms for robot motion and turret rotation.

FTC Team 18340 **January 2022 - July 2024**

Captain *Fremont, CA*

- Designed custom PCBs in KiCad, reducing software testing and validation time by 30%.
- Used top-down assembly design in Onshape to iterate mechanical systems and reduce integration issues.
- Coordinated technical deliverables across subteams and managed a 10-member team with a $10,000 annual budget.

SELECTED TECHNICAL PROJECTS

Digital Logic Design and Verification **Summer 2026**

SystemVerilog, Quartus, ModelSim

- Implemented and simulated RTL modules for shift registers, a barrel shifter, a carry-lookahead adder, and an error-correcting-code encoder and decoder as part of ECE 111 coursework.
- Used testbenches and waveform inspection to evaluate circuit behavior; implemented Mealy and Moore finite-state machines.

MiniTransPac **June 2024 - Present**

Python, Onshape, KiCad, 3D printing, soldering

- Develop an autonomous sailing platform with the long-term goal of crossing the Pacific Ocean.
- Implement Python algorithms for localization and sailing navigation; design a custom hull and wing sail in Onshape.

LEADERSHIP AND RECOGNITION

Fremont Robotics Academy **June 2021 - Present**

Founder and President *Fremont, CA*

- Founded a youth-run nonprofit providing robotics and STEM education; organized classes and FIRST LEGO League tournaments reaching 100+ students across 10 schools.
- Led 100+ volunteers across two countries and 10 annual events, coordinating more than 10,000 volunteer hours since founding.
- Developed community outreach and fundraising campaigns that secured $15,000 in grants and donations.

FIRST Dean's List Finalist **April 2023**

- Recognized as one of 200 worldwide finalists from 85,000 students for leadership and technical expertise.
- Selected as one of four finalists from Northern California.

TECHNICAL SKILLS

Programming: C/C++, Python, Java, SystemVerilog

Circuit and digital design: EasyEDA, KiCad, LTspice, MATLAB, Quartus, ModelSim

Mechanical design and fabrication: Onshape, Fusion 360, SolidWorks, 3D printing, soldering